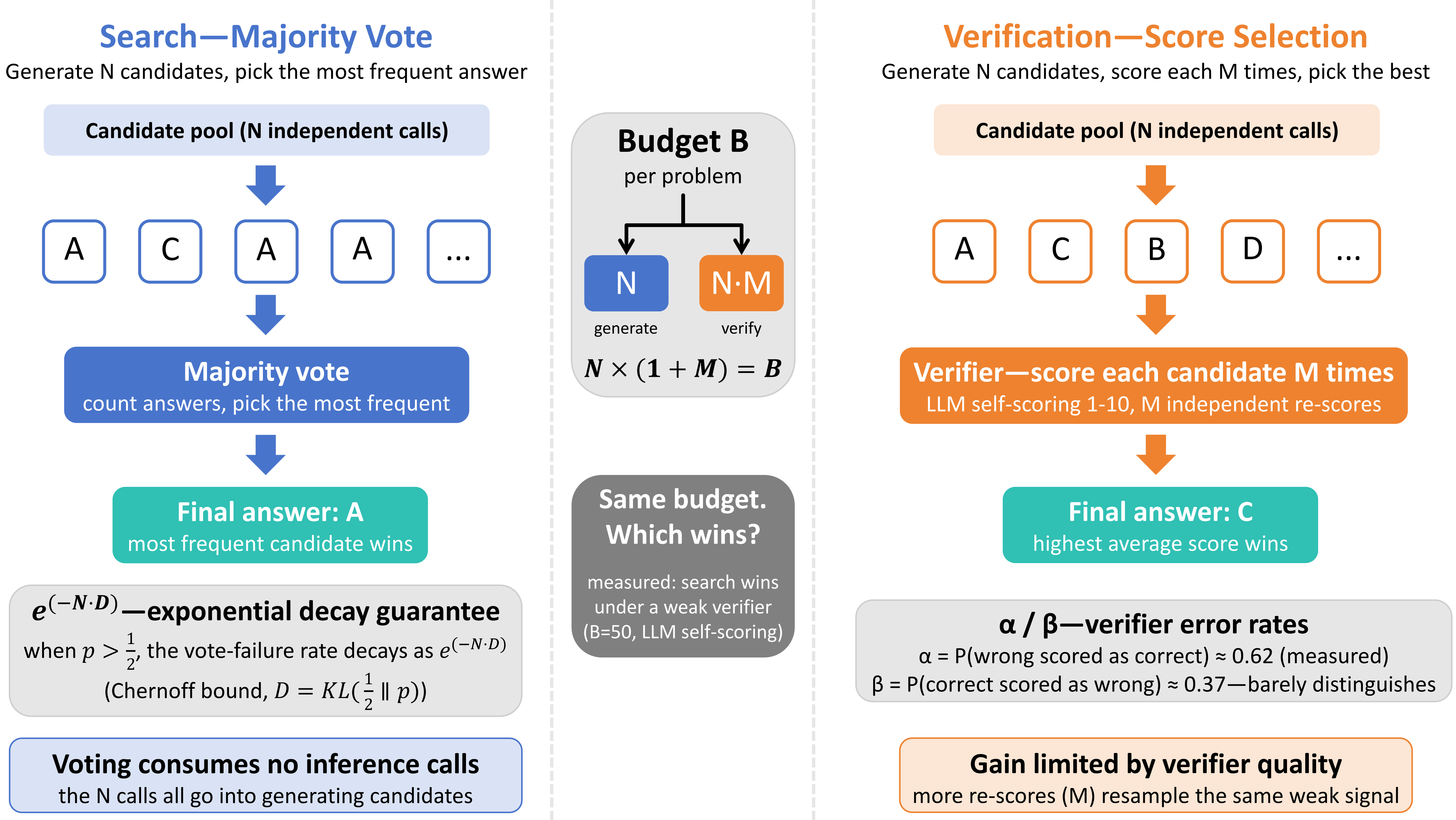
Search—Majority Vote
Generate N candidates, pick the most frequent answer
Candidate pool (N independent calls)
A
C
A
A
...
Majority vote
count answers, pick the most frequent
Final answer: A
most frequent candidate wins
$e^{(-N \cdot D)}$—exponential decay guarantee
when $p > \frac{1}{2}$, the vote-failure rate decays as $e^{(-N \cdot D)}$
(Chernoff bound, $D = KL(\frac{1}{2} \parallel p)$)
Voting consumes no inference calls
the N calls all go into generating candidates
Budget B
per problem
N
N·M
generate
verify
$N \times (1 + M) = B$
Same budget.
Which wins?
measured: search wins
under a weak verifier
(B=50, LLM self-scoring)
Verification—Score Selection
Generate N candidates, score each M times, pick the best
Candidate pool (N independent calls)
A
C
B
D
...
Verifier—score each candidate M times
LLM self-scoring 1-10, M independent re-scores
Final answer: C
highest average score wins
α / β—verifier error rates
α = P(wrong scored as correct) ≈ 0.62 (measured)
β = P(correct scored as wrong) ≈ 0.37—barely distinguishes
Gain limited by verifier quality
more re-scores (M) resample the same weak signal